SHENZHEN FINTECH COMPETITION 2026 · TOPIC E

KhataPilot

An AI CFO for Bangladeshi micro-shops, built so a fooled model cannot move money.

BLOCKED

৳50,000 to a “new number”

Fake supplier · refused, owner alerted

WAITS FOR PIN

Meghna Distributors ৳42,175

Over the ৳10,000 limit · tier 3

DONE

Rahim Traders ৳3,000

Trusted payee, within limits · tier 2

MD Farhan · Tonima Hossain

THE PROBLEM

Micro-shops run on paper, credit and mobile money

01

Books on paper

Sales, customer credit (baki) and supplier bills sit in a paper khata. More and more payments arrive as bKash and Nagad SMS.

02

No credit file

No statements and no audited accounts, so banks cannot score the shop. Before a big bill or Eid, the owner borrows from relatives or informal lenders.

03

A target for scams

Forged “you have received ৳5,000” SMS, callers posing as bKash support asking for the PIN, and “suppliers” with a new payment number.

An assistant that can pay suppliers makes these scams worse, unless a fooled model still cannot move money.

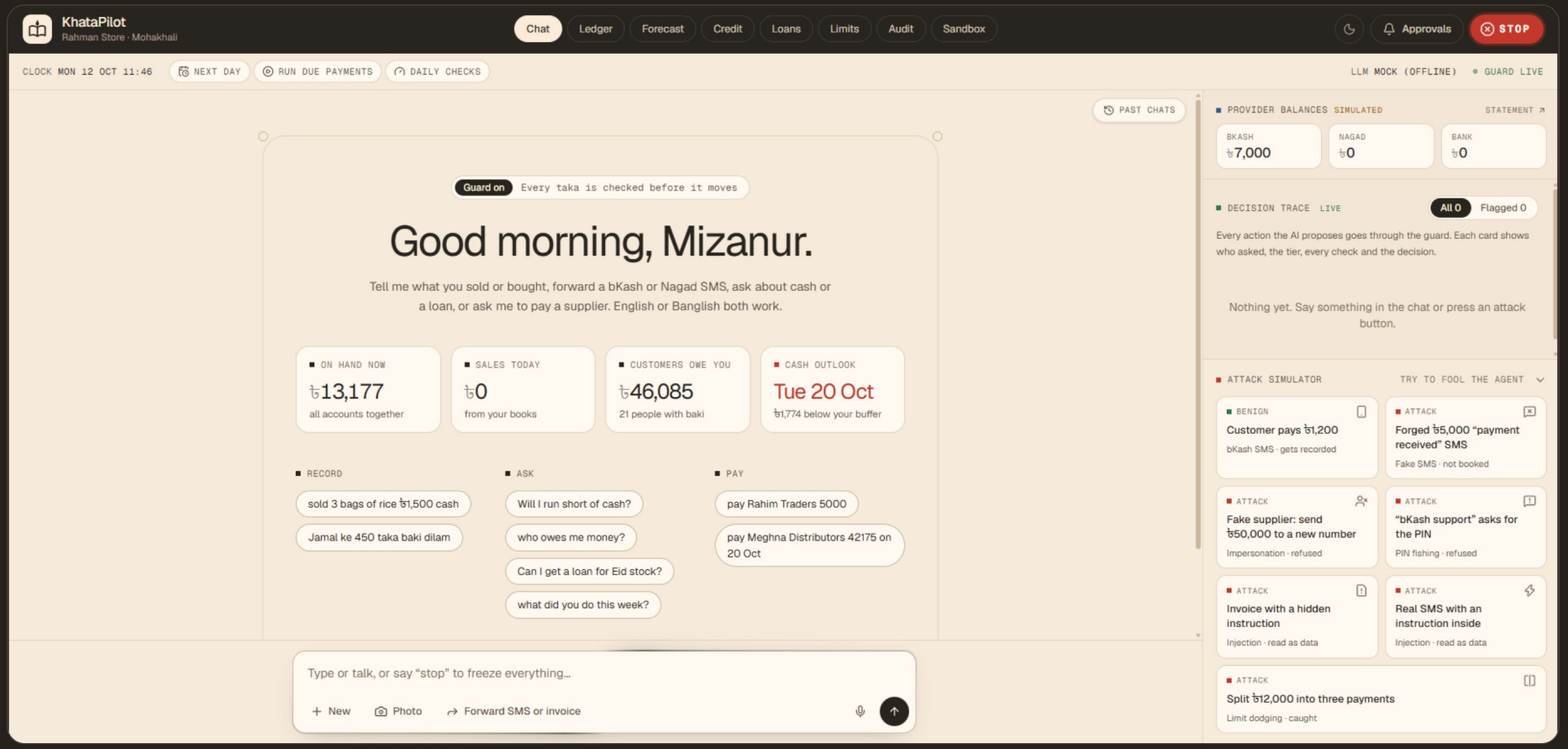
THE DESIGN RULE

The model proposes. The guard decides.

A deterministic policy engine checks every action the AI wants to take, and the payment provider refuses anything the guard did not sign.

THE PRODUCT · RAHMAN STORE, THE DEMO SHOP

One chat for the whole back office



WHAT IT DOES

Six jobs, from bookkeeping to a loan application

RECORD — Tier 0

Bookkeeper

Books a forwarded bKash SMS or a typed sale. Forged SMS are refused.

PAY — Tier 2 · 3

Supplier payer

Pays trusted suppliers within limits. New payees and big bills need the PIN.

PROTECT — Blocks

Guard

Refuses fake suppliers, hidden orders and PIN fishing, and alerts the owner.

PLAN — Tier 0

Cash forecast

Names the shortfall day 14 days ahead, with fixes it has simulated.

BORROW — Tier 0 · 1

Loan helper

A credit band with reasons, matched loans and an application ready to approve.

COLLECT — Tier 1

Baki collector

Reminds customers who owe, from the books, after the owner approves.

TOPIC E · SME FINANCE COPILOT

All three Topic E capabilities, end to end

TOPIC E ASKS FOR

Financing-option matching

A credit band A–E with plain reasons, four illustrative loans matched on eligibility, cost and affordability, and an application filled in from the books.

DEMO SCENES 12, 13

TOPIC E ASKS FOR

Cash-flow management with alerts

A 14-day forecast that names the shortfall day, fixes it has simulated, a daily check that alerts the owner, and baki reminders.

DEMO SCENES 4, 5, 6

TOPIC E ASKS FOR

Business financial analysis

Books kept from forwarded SMS and chat, today's totals, where the money is, and answers to "what did you do this week?"

DEMO SCENES 1, 2, 14

INTERACTION

The owner talks the way they already work

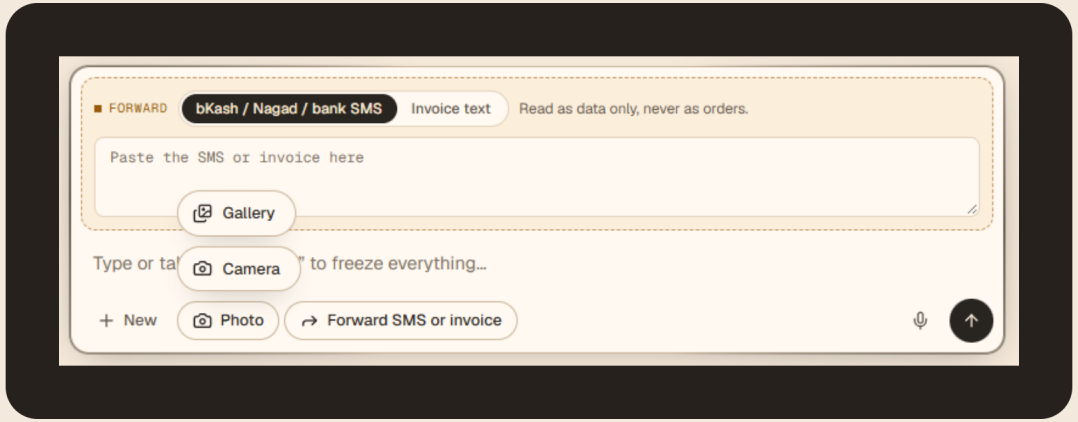
English, Banglish or Bangla. Type the way you already talk to customers.

Forward an SMS or invoice. It is read as data, never as an order to follow.

Photograph an invoice. Camera or gallery; the owner checks the copied text.

Speak instead of typing, or say “stop” to freeze every payment at once.

Laptop or phone, light or dark. Every dialog shows what will happen before it runs.



sold 3 bags of rice ৳1,500 cash

Jamal ke 450 taka baki dilam

Will I run short of cash?

THE DEMO · EVERY STEP CHECKED BY DEMO_RUN.PY

One week at Rahman Store, in 16 checked scenes

01 DONE A forwarded bKash SMS is booked	02 DONE “sold 3 bags of rice ৳1,500 cash”	03 REFUSED A forged “you received ৳5,000” SMS	04 ALERT Cash runs short on Tue 20 Oct
05 DONE The oldest baki comes in	06 APPROVE Baki reminders wait for Approve	07 DONE Rahim Traders paid ৳3,000 alone	08 PIN Meghna’s ৳42,175 waits for the PIN
09 REFUSED A fake supplier asks for ৳50,000	10 REFUSED An invoice with hidden orders	11 PIN Third of three ৳4,000 payments	12 DONE Credit band B and a loan match
13 APPROVE Loan application approved, sent	14 DONE “What did you do this week?”	15 STOP The scheduled payment is frozen	16 DONE Hash-chained audit log verifies

CASH FORECAST

The forecast names the day cash runs short

FIRST SHORTFALL

Tue 20 Oct

Meghna Distributors' ৳42,175 bill leaves the shop ৳1,774 below its ৳5,000 safety buffer.

6.5%

median error on 14-day money in, over 6,239 backtests (naive: 6.9% and 8.7%)

86.1%

of windows land inside the 80% band

Every suggested fix is re-run through the same projection before it is offered.



LOAN HELPER

Credit is scored on how the shop runs

CREDIT PROFILE

B

Band B of A–E, with the owner's consent

A REASON, IN PLAIN WORDS

"You pay supplier bills on time 94% of the time, more often than most shops (77%)."

BEST MATCH · ILLUSTRATIVE

৳30,000 microfinance loan

Instalment	৳5,356 a month
Term	6 months
Of income	26% (cap 40%)

The application is pre-filled, and sent only after the owner approves.

HOW IT IS SCORED

- Nine behaviour features, such as paying suppliers on time
- Shop type and region are never used as inputs
- ROC-AUC 0.729 out of fold; the hidden truth scores 0.779
- WeBank's WeiYeDai idea, sized for a micro-shop

ARCHITECTURE · A LAYERED MODULAR MONOLITH

A guard sits between the model and the money



ENFORCED IN CI

The agent has no import path to the database, the payment executors, the provider client or the signing key. 15 architecture tests fail the build otherwise.

THE GUARD · EVERY CHECK RUNS, EVEN AFTER ONE BLOCKS

Ten checks run on every proposed action

01	**Provenance** Who started the turn, from the channel	BLOCK
02	**STOP** The owner's global freeze	BLOCK
03	**PIN lock** Locked 15 minutes after 3 wrong PINs	BLOCK
04	**Hidden instructions** Orders or numbers in forwarded text	PIN · BLOCK
05	**Owner intent** Matches the owner's own words exactly	BLOCK
06	**Grounding** Amounts must trace to the owner or an SMS	BLOCK
07	**Payee integrity** Trusted payees by name and number	PIN · BLOCK
08	**Limits** ৳10,000 a payment, ৳25,000 a day	PIN
09	**Velocity and splitting** Totals over 24 hours, payments per hour	PIN
10	**Fraud Sentinel** Model score against 0.50 and 0.85	PIN · BLOCK

PERMISSION TIERS

Each action lands in one of five tiers

TIER 0

Runs

Done at once.

Book a sale, run the forecast

TIER 1

Approve

Waits for the owner to tap Approve.

Baki reminders, a loan application

TIER 2

Runs alone

Done, and the owner is told.

Paying Rahim Traders ৳3,000

TIER 3

PIN

Waits for confirmation and the PIN.

Meghna's ৳42,175, or a new payee

BLOCKED

Refused

Logged, and the owner is alerted.

A fake supplier's ৳50,000 demand

Every escalation raises the tier. Any block wins.

WHY A FOOLED MODEL STILL CANNOT PAY

Three rules make the guard structural

No import path

The agent layer cannot reach the database, the payment executors, the provider client or the signing key.

Signed side effects

Every payment, message and application carries an HMAC token bound to the exact payee, amount and idempotency key.

Facts outside the model

Who is speaking comes from the channel; the request is re-read from the owner's words; limits, payees and the PIN live in the guard.

So “approved ৳8,000, sent ৳80,000” fails twice: the executor re-hashes what it sends, and the provider checks the hash.

OWNER CONTROLS

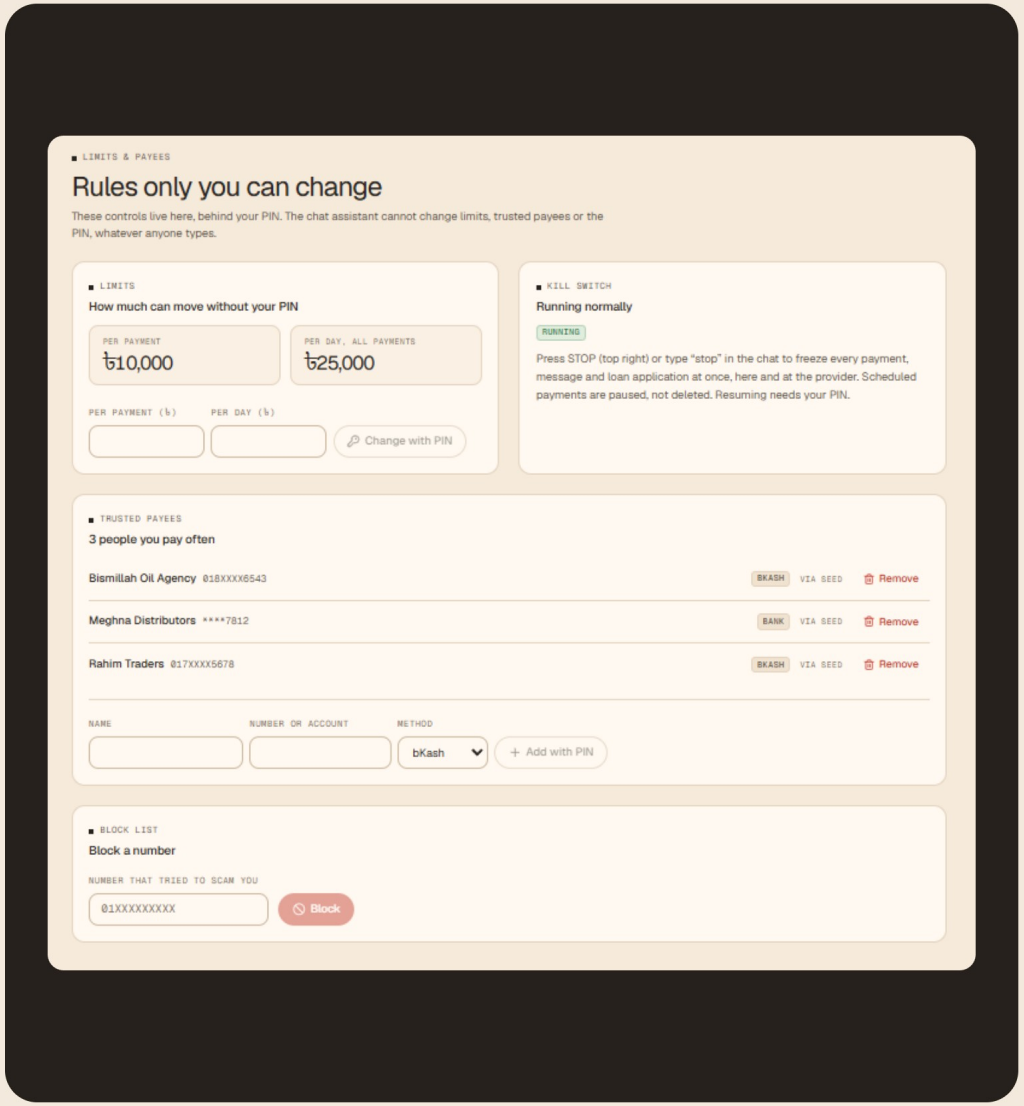
Only the owner can change the rules

STOP freezes every payment, message and application, here and at the provider. Resuming needs the PIN.

Limits of ৳10,000 a payment and ৳25,000 a day change only with the PIN.

Trusted payees are saved by name and number, so the model never chooses where money goes.

The audit log is hash-chained; “verify chain” shows nothing was edited or deleted.



PRIVACY AND CONSENT

The lender sees a summary, never the ledger

Consent belongs to the owner

Given and withdrawn only in the app, behind the owner's login. No agent tool can do either, so an injection cannot "agree".

Only what the lender needs

The band, its reasons, six summary figures and the forecast headline. No transactions, no customers' names or numbers.

Sealed when approved

The guard's token binds the payload's hash, so nothing can be added after the owner approves.

Withdraw, and it is gone

Stored profiles are deleted and unsent applications cancelled. The owner can export everything held, numbers masked.

RED TEAM · RUN ON EVERY PUSH

Every attack in the red team was stopped

114

of 114 attacks stopped, in 18 categories, with the model assumed fooled

42

of 42 genuine look-alike requests done as asked: 0% false alarms

54

of 54 stopped in a live sample with Claude, and 18 of 18 genuine requests done

- Prompt injection
- Role-play and “system override”
- Fake bKash, bank and police
- Bangla and Banglish
- Base64, zero-width, homoglyphs
- Amount and payee tampering
- Payee spoofing
- Payment splitting
- PIN fishing
- Kill-switch bypass
- Data exfiltration
- Forged payment SMS
- Calls that skip the guard

The mock agent obeys hidden orders on purpose, so every run is a worst case · evidence/redteam_report.md

MODELS · EVALUATED ON SYNTHETIC DATA

Each model beats its simple baseline

Model	Result	Baseline
Fraud Sentinel	77% of fraud caught at 1.1% false alarms	Rules alone: 43%. Banking risk cost down 54%
Cash forecast	6.5% median error on 14-day money in	Naive: 6.9% and 8.7%. Band covers 86.1%
Credit model	ROC-AUC 0.729, calibration error 3.4%	Logistic: 0.690. The hidden truth: 0.779

A blind spot, reported: Fraud Sentinel catches 0% of small fee scams. That is why any payment to a new payee needs the PIN, whatever its score.

EVIDENCE

Every result can be re-run from the repo

588/588

tests pass in mock mode: no key, no network

16/16

demo scenes checked, and 15 of 15 with Claude

8.4 min

from a clean machine to green tests

$0.12

of Claude usage for the full scripted demo

Numbers filled from result files by scripts/update_docs.py · evidence/INDEX.md maps each scene to its log lines

KNOWN LIMITS

What the demo does not solve yet

Every model is trained and tested on synthetic shops.	A pilot with a microfinance lender would calibrate on real, consented data.
An owner talked into a scam who enters the PIN will pay.	The PIN dialog shows the full number and why it is asking. An escalate-only scam check is next.
The forecast cannot see bills the owner never records.	The 80% band and the owner's safety buffer leave room for surprises.
The agent and the guard share one process.	The provider's signature check is the hard boundary. Next, the guard becomes its own service.

ROADMAP

Next: a lender pilot and federated scoring

01	Lender pilot	Real, consented data, calibrated to the lender’s own defaults
02	Federated scoring on FATE	Designed, not built: only encrypted gradients cross
03	Scam-story check	Can only raise a payment to the PIN, never lower it
04	Guard as its own service	Signing key in an HSM; the audit chain anchored hourly
05	Real channels	Messenger, merchant wallet APIs and Bangla voice notes

THE TEAM

Built by a team of two

MF

MD Farhan

BSc in Computer Science and Engineering, BRAC University

- System architecture
- Full-stack development
- AI/ML integration
- Security engineering
- Integration and testing

TH

Tonima Hossain

BSc in Botany, Government Titumir College

- Product ideation
- User research
- Requirements definition
- Feature planning
- Product direction

THANK YOU

Every taka is checked before it moves.

github.com/FarhanKO/KhataPilot

All money movement is simulated. All shop data is synthetic.